ENGINEERING PORTFOLIO

ANDREW MARION

MECHANICAL ENGINEERING · AEROSPACE

Junior in Mechanical Engineering with an Aerospace minor at Kennesaw State University. My work splits between hands-on hardware (rocket propulsion, autonomous aircraft, ground support equipment) and CFD research validated against wind-tunnel and experimental data. Currently a UAV Flight Control Intern at NASA Glenn Research Center and Vice President of the Kennesaw Rocketry Team.



3.81
GPA · KSU

May '27
GRADUATION

4
SCITECH 2027 PAPERS ACCEPTED

140+
CFD SIMULATIONS RUN

CONTACT

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EDUCATION

Kennesaw State University
B.S. Mechanical Engineering, Aerospace Engineering Minor
Anticipated May 2027 · U.S. Citizen

COURSEWORK

Aircraft Design, CAE, Aerodynamics, Fluid Dynamics, Heat Transfer

CONTENTS

Internships · Build Projects · Research · Publications · Skills

01 INDUSTRY INTERNSHIPS

NASA GLENN RESEARCH CENTER — CLEVELAND, OH AUG – DEC 2026

INTERNSHIP

UAV FLIGHT CONTROL INTERN

Diagnostics and Controls Branch · Distributed electric propulsion UAV based on NASA's SUSAN Electrofan



- Implementing custom MATLAB/Simulink flight control algorithms on Pixhawk flight computers running PX4 for a 12 ft wingspan distributed electric propulsion UAV with 16 electric fans and thrust vectoring
- Developing control reallocation algorithms in MATLAB that redistribute thrust commands across the remaining fans after a fan or thrust-vectoring failure, tested across 8,000 fan-failure cases with Monte Carlo simulation
- Modeling flight dynamics and powertrain in MATLAB/Simulink and X-Plane, and performing system identification on flight test data with Kalman filtering and airspeed, IMU, and GPS sensor fusion
- Integrating algorithms into PX4 in a Linux environment; invited to return in spring 2027 to continue the project

MATLAB/Simulink PX4 Pixhawk Kalman Filters System ID X-Plane



TEXTRON GSE — CARTERSVILLE, GA JUN – AUG 2026

INTERNSHIP

MECHANICAL DESIGN ENGINEER INTERN

Ground support equipment electrification

- Designed assemblies in Siemens NX and produced build-ready drawings, including a hinged door for the MX4 electric baggage tractor and access doors for the 660 belt loader refresh, with hands-on sheet metal design and prototyping
- Applied GD&T and tolerance stack-up analysis alongside DFA/DFM reviews to validate manufacturability before production release
- Ran fit checks and physical prototype testing, including parking-brake sensor location testing and access-door motion verification

Siemens NX GD&T Sheet Metal DFA/DFM

02 BUILD PROJECTS



AERIAL ROBOTICS COMPETITION TEAM SEP 2023 – APR 2025

TEAM LEAD

POWERED AUTONOMOUS DELIVERY AIRCRAFT (PADA)

SAE AERO Design — Team Lead, 20+ members

- Led the full design, manufacturing, and flight-test lifecycle of a carrier-launched delivery aircraft, from CAD through fabrication and integration
- Manufactured a carbon fiber mold and performed a wet layup to construct the fuselage
- Built and flew a custom flying wing that detached mid-flight from a larger carrier aircraft and autonomously navigated to landing coordinates it detected in flight
- Configured and tuned a Matek H743 flight computer for stable flight on an inherently unstable airframe



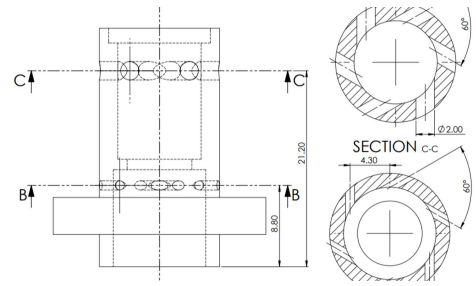
KENNESAW ROCKETRY TEAM DEC 2025 – PRESENT

OPERATIONS / CFD

IREC COMPETITION ROCKET

Vice President — Team Operations Lead

- Led team operations for the Intercollegiate Rocket Engineering Competition build across Rocket Design, Avionics, and Payload subteams
- Optimized rocket body and camera mount geometry against drag through high-fidelity CFD analysis
- Assisted with fiberglass construction and resolved fin-to-airframe gap issues through iterative SolidWorks corrections
- 3D printed the camera mount and payload components; standardized build documentation



KENNESAW ROCKETRY TEAM DEC 2025 – PRESENT

LIQUID PROPULSION

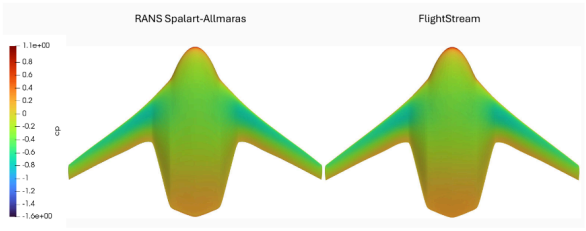
BIPROPELLANT ROCKET ENGINE

Vice President; previously Liquid Propulsion Lead

- Designing and sizing an N_2O / ethanol bipropellant engine targeting 250 lbf thrust at an O/F ratio of 4.8: thrust equations, injector pressure drop, chamber pressure, and mass-flow calculations
- Designing an injector cold-flow test stand around a pressurized nitrogen supply (~150–220 psi)
- Reviewed and refined a coaxial swirl injector assembly through SolidWorks design checks ahead of fabrication

03 CFD & AERODYNAMICS RESEARCH

KSU Advanced Aerodynamics and Propulsion Lab: simulation work validated against wind-tunnel and experimental data, published at AIAA conferences.

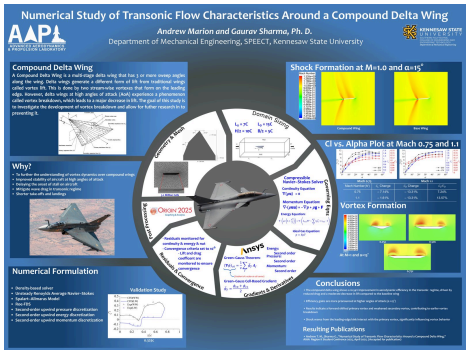


JAN 2026 – PRESENT 1ST AUTHOR ×2

BLENDED WING BODY VS. RANS CFD

Siemens FlightStream benchmarking

- Benchmarking FlightStream against RANS-based CFD for a blended wing body (BlendedNet++ dataset, Geometry 227) in collaboration with Siemens
- Assessing FlightStream's potential to speed up early-stage aircraft concept simulation
- Two papers accepted as first-author work at AIAA SciTech 2027

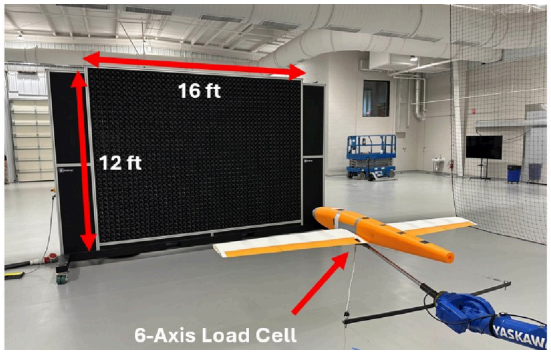


SEP 2023 – PRESENT 1ST AUTHOR

COMPOUND DELTA WING CFD

Vortex lift & transonic performance

- Executed 140+ CFD simulations in ANSYS Fluent across velocities and angles of attack, improving lift-to-drag efficiency from 7.24% to 13.57% in the transonic regime
- Investigated vortex lift modeling through trailing-edge boundary condition treatment
- First-authored paper at the 2025 AIAA Region II Student Conference

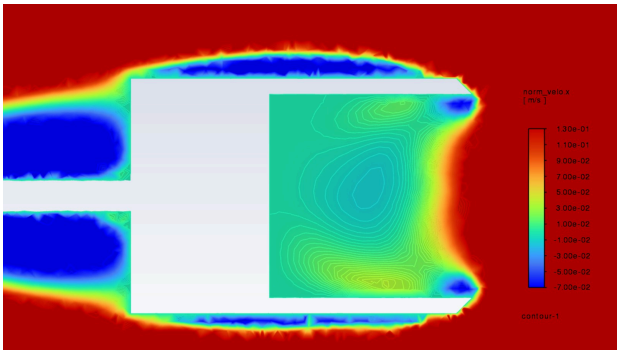


MAR 2025 – PRESENT CO-AUTHOR

CONTROL SURFACE VALIDATION — "BIRD" UAV

KSU × Oklahoma State S2FAR Lab

- Validated FlightStream aileron-deflection load predictions against high-fidelity CFD and WindShape wind-tunnel / load-cell testing on a Group-1 fixed-wing UAV
- Created aerodynamic models informing initial flight conditions for closed-loop roll-axis control validation on a Pixhawk Cube Orange+
- Co-authored findings at AIAA SciTech 2026



IN PROGRESS DESIGN PHASE

FORWARD-FACING CAVITY PIV REPLICATION

URANS CFD vs. experimental PIV

- Running URANS CFD to replicate a forward-facing cylindrical cavity PIV experiment, working through timestep selection and Rossiter mode analysis
- Coordinating validation of vorticity and velocity distribution against experiment

04 PUBLICATIONS & PRESENTATIONS

JAN 2027 ACCEPTED
Aerodynamic Characterization of a Blended Wing Body Using a Surface Vorticity Solver
Marion, A., Sharma, G., Sirak, E., and Shahjahan, S. · AIAA SciTech 2027 Forum

JAN 2027 ACCEPTED
Modeling Vortex-Dominated Flow over a Delta Wing Using a Surface Vorticity Solver
Marion, A., Sharma, G., Sirak, E., and Shahjahan, S. · AIAA SciTech 2027 Forum

JAN 2027 ACCEPTED
Aerodynamic Analysis of a Blended Wing Body Using the Lattice Boltzmann Approach
Mehar, P., Marion, A., Sharma, G., Muller, M., and Gaedtke, M. · AIAA SciTech 2027 Forum

JAN 2027 ACCEPTED
Cross-Validation of a Mid-Fidelity Fixed-Wing Aerodynamic Model for Control Law Development with Flight Testing
Atkinson, Z., Comer, A., Marion, A., and Sharma, G. · AIAA SciTech 2027 Forum

JAN 2026 CONFERENCE PAPER
Multi-Fidelity Validation of Control Surface Deflection Predictions Using FlightStream
Atkinson, Z., Comer, A., Marion, A., Sharma, G., Shahjahan, S., and Wilson, T. · AIAA SciTech 2026 Forum · doi.org/10.2514/6.2026-2810

MAY 2025 INVITED TALK
Study of Compound Delta Wings and Kennesaw State Undergraduate Research Experience
Marion, A. · Presented to the President's Community Advisory Board

APR 2025 CONFERENCE PAPER
Numerical Study of Transonic Flow Characteristics Around a Compound Delta Wing
Marion, A., and Sharma, G. · 2025 AIAA Region II Student Conference · doi.org/10.2514/6.2025-97857

MAY 2024 POSTER
Numerical Analysis of Vortex Breakdown Over a Compound Delta Wing
Marion, A., and Sharma, G. · Symposium of Student Scholars

MAR 2026 FEATURE STORY
Engineering Student Uses Computer Simulations to Shape the Future of High-Speed Flights
Kennesaw State University News

05 TECHNICAL SKILLS

CFD SOFTWARE
ANSYS Fluent, FlightStream, SolidWorks Flow Simulation, OpenVSP, ParaView

CAD SOFTWARE
SolidWorks, Siemens NX, AutoCAD, ANSYS SpaceClaim

PROGRAMMING / AUTOPILOT
MATLAB/Simulink, Python, R, LaTeX, PX4, ArduPilot, Pixhawk

FABRICATION & SHOP
Sheet metal prototyping, 3D printing, laser cutting, composite layup, lathe/mill, CNC machining, foam cutting

DESIGN PRACTICE
GD&T, tolerance stack-up, DFA/DFM reviews, fit checks, build documentation

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06 HIKING



From the Appalachian Trail in Georgia to Exit Glacier in Alaska. An interactive map of all 15 stops is on my website.